PURE PROJECTIVE RESOLUTIONS EXIST AND LIFT MAPS UNIQUELY UP TO HOMOTOPY

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file adequate.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof writes F_0 once where the term of the complex named just above is P_0, and writes d_{F,1} once where the differential of P is meant (d_{P,1}). Both are preserved verbatim and recorded here rather than corrected.

Editorial note. The counted spans carry navigation stubs for the references leaving them; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- Every module has a pure projective resolution. Same chapter, Lemma lemma-pure-projective, invoked at proof lines 2829-2830. Gives part (1).
- A universally exact resolution splits into universally exact short exact sequences of syzygies. Same chapter, Lemma lemma-split-universally-exact-sequence, invoked at proof lines 2831-2835. Provides the short exact sequences used by both inductions.

Source reference. Original lemma 06ZZ: sources/stacks-adequate/pinned-original-source.tex, source lines 2683–2683. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0701: sources/stacks-adequate/pinned-original-source.tex, source lines 2760–2760. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let A be a ring.*

(1) *Any A-module has a pure projective resolution.*

Let $M \to N$ be a map of A-modules. Let $P_\bullet \to M$ be a pure projective resolution and let $N_\bullet \to N$ be a universally exact resolution.

(2) *There exists a map of complexes $P_\bullet \to N_\bullet$ inducing the given map*

$$M = \mathrm{Coker}(P_1 \to P_0) \to \mathrm{Coker}(N_1 \to N_0) = N$$

(3) *two maps $\alpha, \beta : P_\bullet \to N_\bullet$ inducing the same map $M \to N$ are homotopic.*

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Proof. Part (1) follows immediately from Lemma 06ZZ. Before we prove (2) and (3) note that by Lemma 0701 we can split the universally exact complex $N_\bullet \to N \to 0$ into universally exact short exact sequences $0 \to K_0 \to N_0 \to N \to 0$ and $0 \to K_i \to N_i \to K_{i-1} \to 0$.

Proof of (2). Because P_0 is pure projective we can find a map $P_0 \to N_0$ lifting the map $P_0 \to M \to N$. We obtain an induced map $P_1 \to F_0 \to N_0$ which ends up in K_0. Since P_1 is pure projective we may lift this to a map $P_1 \to N_1$. This in turn induces a map $P_2 \to P_1 \to N_1$ which maps to zero into N_0, i.e., into K_1. Hence we may lift to get a map $P_2 \to N_2$. Repeat.

Proof of (3). To show that α, β are homotopic it suffices to show the difference $\gamma = \alpha - \beta$ is homotopic to zero. Note that the image of $\gamma_0 : P_0 \to N_0$ is contained in K_0. Hence we may lift γ_0 to a map $h_0 : P_0 \to N_1$. Consider the map $\gamma_1' = \gamma_1 - h_0 \circ d_{P,1} : P_1 \to N_1$. By our choice of h_0 we see that the image of γ_1' is contained in K_1. Since P_1 is pure projective may lift γ_1' to a map $h_1 : P_1 \to N_2$. At this point we have $\gamma_1 = h_0 \circ d_{F,1} + d_{N,2} \circ h_1$. Repeat. $\square$